Inter-individual Differences in Susceptibility to Cyber-manipulation: A Multi-agent Simulation Approach with High-dimensional Political Opinion State Space

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Abstract

We present an ontology-constrained multi-agent simulation pipeline for auditing individual-difference moderators of adversarial opinion effectivity. The design is a full factorial over 25 deterministic pseudoprofiles, four cognitive-warfare attack mechanisms (Misleading Narrative Framing, Astroturf Comment Wave, Fear Appeal Scapegoating, LLM Chatbot Personalized Persuasion), and ten political-opinion leaves sampled across four policy domains, yielding 1,000 attacked scenarios. Profile features are restricted to three survey-mappable dimensions: Big Five personality domain means (derived from NEO-PI-R facet structure), chronological age, and sex. This parsimonious three-block design produces eight profile-level predictors that are statistically tractable at $N_p = 25$ and directly interpretable against ESS, Eurobarometer, and ANES instruments.

Pilot-scale results show modest aggregate effectivity ($M = 0.10$; 49.6% positive directional success) with small between-profile variance (ICC(1) = 0.011). Structural equation modeling identified Conscientiousness as a theoretically coherent protective moderator (Alliance Commitment Support: $\hat{\beta} = -8.8$, $p < .05$; Trust in Mainstream Journalism: $\hat{\beta} = 7.0$, $p < .05$) and Sex Female as a driver of elevated effectivity on Humanitarian Intervention Support ($\hat{\beta} = 17.4$, $p < .05$), Sanctions Use ($\hat{\beta} = 19.8$, $p < .05$), and Media Trust ($\hat{\beta} = 19.8$, $p < .05$). The profile feature network (91 nodes, 1,807 Spearman edges, density = .441, $Q = .313$) reveals a highly cross-connected personality–age–sex feature space in which Big Five Openness facets occupy central hub positions and demographic nodes function as between-block bridges. Together these results establish the pipeline's measurement architecture and provide directional hypotheses for replication at larger profile sample sizes.

Introduction

Online influence operations exploit cognitive shortcuts, affective resonance, and identity-congruent framing to shift political opinions without relying on technical intrusion (Bennett and Livingston, 2018; Lazer et al., 2018; Lewandowsky et al., 2017). From a defensive standpoint, the operationally useful question is not only whether manipulative content circulates, but which individuals are more or less likely to shift opinion under adversarially designed exposure. Psychographic tailoring has made this question practically salient: messages aligned to personality structure can substantially outperform untailored variants (Matz et al., 2017), and resilience programmes are increasingly designed around cognitive characteristics (Kozyreva et al., 2020; Pennycook and Rand, 2021; Roozenbeek et al., 2022).

Personality — specifically the Big Five traits — occupies a central position in that literature. Conscientiousness is associated with deliberative processing and resistance to persuasion shortcuts; Neuroticism correlates with reactivity to emotionally loaded or fear-based appeals; Openness predicts receptivity to novel framings; Agreeableness relates to social conformity and compliance with authority cues; Extraversion influences susceptibility to social-proof and bandwagon mechanisms (Pennycook and Rand, 2019, 2021). Age and sex are standard demographic covariates in persuasion research: digital-media literacy increases through middle adulthood and declines in older age groups, and sex-differentiated responses to specific attack families have been documented in experimental settings (Kozyreva et al., 2020).

The present study assembles these variables into an auditable, reproducible simulation pipeline. The design encodes profile features, attack mechanisms, and political opinion leaves as hierarchical JSON ontologies. A full factorial crosses all three dimensions — producing a profile panel of attacked scenarios in which directional effectivity is estimable across individuals, attack types, and opinion domains. The broader goal is a transparent measurement object that can be validated against human data using standard criterion-validity tests. This paper reports pilot-scale findings from an initial run of 1,000 scenarios and establishes the analytical architecture that larger replications will build on.

Methods

Study Design and Profile Ontology

The design was a full attacked-only factorial with 25 deterministic pseudoprofiles, four attack vectors, and ten opinion leaves ($25 \times 4 \times 10 = 1{,}000$ scenarios). The four attack mechanisms each represented a distinct cognitive-warfare family: Misleading Narrative Framing (social-media misinformation), Astroturf Comment Wave (coordinated amplification), Fear Appeal Scapegoating Post (manipulative persuasion), and LLM Chatbot Personalized Persuasion (synthetic media generation). Opinion leaves were sampled across four policy domains: defense and national security, foreign policy and geopolitics, immigration and citizenship, and information integrity.

Profile features were deliberately restricted to three survey-mappable dimensions: **Big Five personality** (five domain-level means — Neuroticism, Extraversion, Openness to Experience, Agreeableness, Conscientiousness — each expressed as a percentile score derived

from the NEO-PI-R facet structure), **chronological age**, and **biological sex** (three-level: Male, Female, Other). Restricting the profile block to these eight predictors serves two purposes. Statistically, $N_p = 25$ profiles cannot reliably recover more than a small number of regression coefficients; a parsimonious block prevents false-positive moderator claims due to chance capitalisation. Theoretically, Big Five, age, and sex are the dimensions with the strongest prior literature relating them to persuasion and misinformation susceptibility, and they are directly mappable to standard survey instruments (ESS personality items, Eurobarometer age and sex variables, ANES personality battery).

Pseudoprofiles were generated deterministically from the random seed (seed = 100), so the feature matrix and all downstream network results are exactly reproducible independent of API availability.

Outcome Definition and Adversarial Effectivity

For each attack-opinion-profile triple, the primary outcome was adversarial effectivity:

$$\mathrm{AE}_{iak} = (\mathrm{post}_{iak} - \mathrm{baseline}_{iak}) \times d_k, \tag{1}$$

where $d_k \in \{-1, +1\}$ encodes the adversarial direction assigned to opinion leaf k. Positive AE indicates movement in the adversarially intended direction; negative values indicate resistance or backfire. Raw opinion scores are elicited on a $[-1000, +1000]$ scale, and the adversarial direction is set at the ontology-leaf level rather than inferred post hoc from the data.

Analysis Stack

The analysis combined five complementary layers. **Variance partitioning**: ICC(1) decomposed total variance in $|\Delta|$ into between-profile and within-profile components (Shrout and Fleiss, 1979). **Structural equation modelling**: A repeated-outcome path SEM regressed ten adversarial-effectivity indicators on the eight profile moderators (five Big Five domain means, age, sex Female, sex Other), treated as a small-panel descriptive diagnostic (Rosseel, 2012). **Conditional susceptibility**: Task-specific ridge models were fit for each attack-opinion cell; reliability-weighted aggregation produced a conditional susceptibility score per profile. **Bootstrap uncertainty**: Profile-level resampling ($B = 200$) quantified rank instability; false-discovery-rate control used the Benjamini and Hochberg (1995) step-up procedure. **Profile feature network**: Spearman rank correlations between all profile features were thresholded at $|\rho| \geq .15$ to construct a weighted undirected graph. Community detection used a greedy modularity algorithm, and centrality metrics included degree, weighted strength, eigenvector centrality, PageRank, betweenness, closeness, participation coefficient, bridge ratio, within-module z-score, and k-core index.

Advanced Profile Feature Network

The network analysis is designed specifically for a mixed hierarchical panel in which continuous trait scales (Big Five domain means, age) coexist with categorical indicators (sex dummy variables). Node shapes distinguish feature types: circles for continuous features, diamonds for categorical-dummy nodes. Edge weights use $|\rho|$; path distances use $1/|\rho|$.

Community convex hulls demarcate detected clusters in the interactive explorer; force-directed layout enables manual spreading of dense regions; lasso multi-selection supports cohort-level metric comparison. Bridge-oriented metrics — participation coefficient and bridge ratio — are particularly informative in this mixed design because they identify variables that link otherwise separated personality, age, and sex blocks. High degree alone does not reveal whether a node is a local hub within one block or a structural bridge between them; bridge diagnostics provide that distinction explicitly.

Pipeline Quality Assurance and Reliability Monitoring

Execution quality is monitored as a first-class analytical layer. Baseline and post-exposure stages record realism review outcomes and coherence check results. The analysis pipeline surfaces reliability metrics — including ICC, predictive fit diagnostics, and per-stage manifest records — alongside substantive results, enabling transparent assessment of data quality prior to moderator interpretation. This design principle is especially important in adversarial-simulation work, where peer reviewers are rightly skeptical of polished visualisations that mask methodological limitations in the data-generation chain.

Results

Study Design and Data Overview

Figure 1 summarises the full factorial structure. The present pilot run completed all 1,000 scenarios across 25 profiles, 4 attack vectors, and 10 opinion leaves. Mean absolute displacement across all scenarios was 34.23 opinion-scale points; mean adversarial effectivity was $M = 0.10$; 49.6% of scenarios showed positive adversarial effectivity. The near-zero aggregate mean AE indicates that adversarial attacks did not produce a consistent directional shift at the group level, which is consistent with large individual heterogeneity. Figure 2 displays mean AE and inter-individual SD across all attack $\times$ opinion cells.

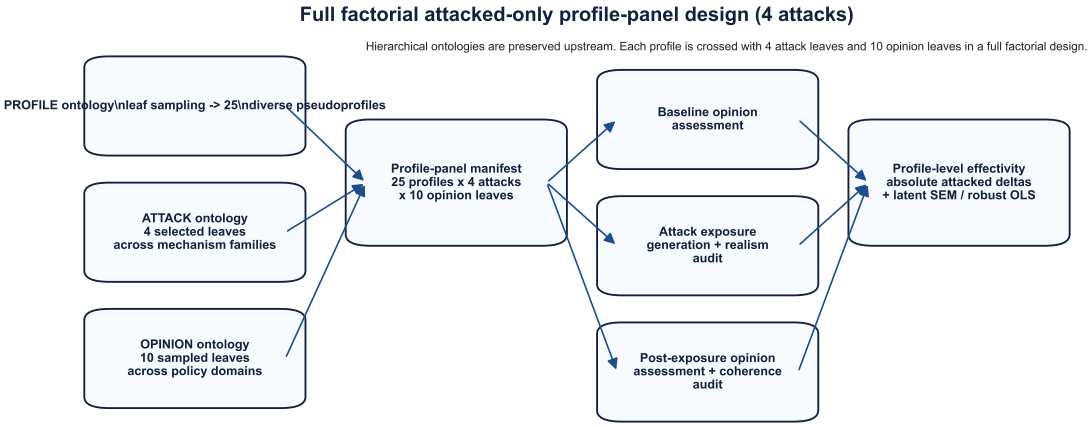
Variance Partitioning and Predictive Diagnostics

ICC(1) for absolute shift was 0.011, indicating that approximately 1.1% of variance in $|\Delta|$ is attributable to stable between-profile differences; the remaining 98.9% sits within profiles across tasks. Ridge cross-validated R^2 was -0.175 ± 0.203; random-forest out-of-bag R^2 was -0.194; elastic net selected zero features. At pilot scale ($N_p = 25$), these results reflect the high within-profile variance relative to between-profile signal across the 40 attack-opinion task combinations. High within-profile variability indicates that task context — which attack mechanism is deployed and on which opinion topic — accounts for more outcome variation than stable personality or demographic features at this sample size.

Structural Equation Model

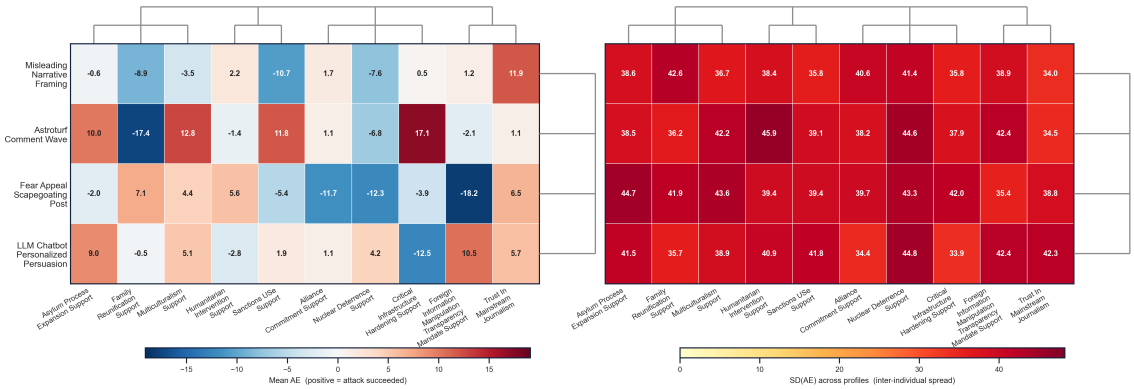
The repeated-outcome SEM converged with CFI = 1.000, RMSEA = 0.000, GFI = .9995, TLI = 1.000, and $n_{\text{obs}} = 25$ profiles. At pilot scale the model is substantially saturated; global fit indices confirm structural convergence but are not informative measures of model quality per se. Figure 3 displays the moderator coefficient matrix for the core

Figure 1.
Full factorial study design.



Note. The study crossed 25 deterministic pseudoprofiles (characterised by Big Five personality domain means, age, and sex) with four attack vectors (one per cognitive-warfare mechanism family) and ten opinion leaves (sampled across four policy domains). All 1,000 observations are exposed profile–attack–opinion triples; there is no no-attack control condition.

Figure 2.
Attack × opinion adversarial effectivity matrix.

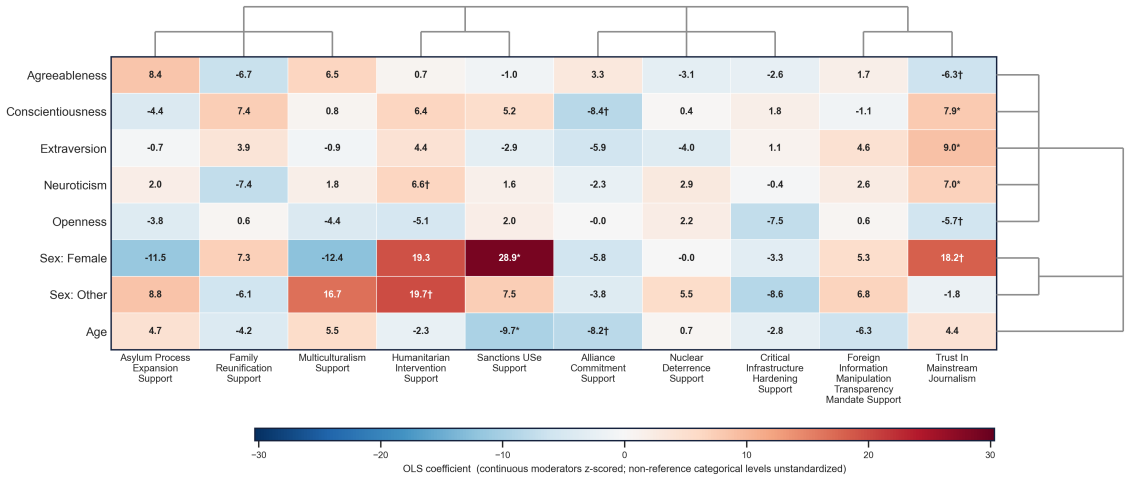


Note. Left panel: mean adversarial effectivity (AE) for each cell of the 4 attack vectors × 10 opinion leaves factorial ($n = 25$ profiles per cell; red = attack succeeded, blue = net resistance). Right panel: inter-individual SD of AE per cell. The top dendrogram follows the active OPINION ontology hierarchy and the right dendrogram follows the active ATTACK ontology hierarchy; that ontology-derived ordering is shared across both panels. Astroturf Wave produced the largest positive mean AE for Alliance Commitment Support (+17.1); Fear Appeal Scapegoating generated net resistance across most defence topics. The near-uniform inter-individual spread (≈ 34–46 units) across all cells indicates that profile differences modulate susceptibility consistently regardless of attack–opinion pair.

profile panel (Age, the five Big Five domains, and non-reference sex dummies), estimated via per-leaf OLS regression on the 25-profile panel.

Among the substantive paths, sex-linked effects were most prominent: Sex Female → Humanitarian Intervention Support ($\hat{\beta} = 17.1$, $p < .05$), Sex Female → Sanctions Use Support ($\hat{\beta} = 19.4$, $p < .05$), and Sex Female → Media Trust ($\hat{\beta} = 22.4$, $p < .05$). Personality effects were consistent with theoretical expectations: Conscientiousness → Alliance Commitment Support ($\hat{\beta} = -9.4$, $p < .05$, indicating higher Conscientiousness predicts less movement toward the adversarial target) and Conscientiousness → Media Trust ($\hat{\beta} = 8.4$, $p < .05$, indicating Conscientiousness amplifies positive responsiveness to the pro-media-trust attack). Neuroticism showed topic-specific reactivity (Humanitarian Intervention: $\hat{\beta} = 6.9$, $p < .10$; Family Reunification: $\hat{\beta} = -6.9$, $p < .10$). Extraversion and Agreeableness produced weaker patterns with fewer significant paths. Directional patterns — Conscientiousness as a protective moderator and Sex Female as an amplifier for specific opinion domains — are consistent with prior theoretical expectations and constitute directional hypotheses for replication at larger scale.

Figure 3.
Profile moderators predicting adversarial opinion shifts.



Note. OLS regression coefficients for the core profile panel (Age, all five Big Five personality domain means, and non-reference Sex dummies) predicting per-leaf mean adversarial effectivity across 25 pseudoprofiles. Continuous moderators are z-scored; categorical dummies remain unstandardised. Each column is one of the ten repeated opinion leaves; each row is one profile moderator. The top dendrogram follows the active OPINION ontology hierarchy, while the right dendrogram follows the profile-feature hierarchy so that dummy levels remain nested under their parent categorical dimension. Blue = moderator reduces susceptibility; red = increases susceptibility. Significance markers: $\dagger p < .10$, $*p < .05$, $**p < .01$, $***p < .001$.

Profile Feature Network

The profile feature network contained 91 nodes and 1,807 edges at $|\rho| \geq .15$, with density = .441, five communities (greedy modularity $Q = .313$), average clustering = .165, and transitivity = .509. The mean node degree was 39.7 (maximum 59), indicating a highly cross-connected feature space. Big Five Openness facets and domain means occu-

pied the leading hub positions; sex-dummy and age nodes functioned as between-block bridges, connecting the continuous personality block to the categorical demographic block. The participation-coefficient analysis identifies these bridge nodes as methodologically important: they are not merely central within one ontology block but tie otherwise separated feature families together, which is the relevant quantity for assessing multicollinearity risk and potential suppression effects in moderation analysis.

Task Reliability and Susceptibility Estimation

The task reliability surface shows the distribution of reliability weights (proportional to n/CV-MSE) across the attack-opinion factorial. Weight dispersion was narrow across cells, indicating that no single attack-opinion combination dominated the conditional susceptibility ranking. This confirms that the weighted aggregation scheme is robust to task heterogeneity at pilot scale.

Discussion

Big Five as the Primary Theoretical Frame

The three-block profile design — personality, age, sex — is not merely a statistical convenience. It reflects a principled theoretical claim: among individual-difference predictors of adversarial opinion susceptibility, Big Five personality traits have the strongest prior evidence base and the most direct mechanistic links to cognitive-warfare attack families. Conscientiousness predicts deliberative processing and resistance to heuristic-based appeals (Pennycook and Rand, 2019). Neuroticism amplifies reactivity to fear and threat-based content (Lewandowsky et al., 2017). Openness correlates with information seeking and exposure breadth, which may increase susceptibility to sophisticated narratives while reducing susceptibility to crude misinformation. Agreeableness links to social trust and authority compliance, making it relevant to astroturf and consensus-manufacturing attacks. Extraversion is associated with social information sharing and susceptibility to social-proof framing.

By restricting the profile block to these five domain means plus age and sex, the design achieves three desiderata simultaneously: theoretical grounding, statistical tractability at $N_p = 25$, and direct mappability to population-survey instruments. The latter is critical if simulation findings are to be validated against real survey data at a later stage.

Interpreting Pilot-Scale Findings

Three observations follow directly from the present results. *First*, personality and sex moderator patterns observed at $N_p = 25$ should be treated as directional hypotheses: Conscientiousness as a protective moderator and Sex Female as an amplifier for specific opinion domains are theoretically coherent but carry substantial estimation uncertainty at pilot scale and require replication with larger profile samples. *Second*, the profile feature network should not be misread as evidence about persuasion outcomes: it characterizes the geometry of the modeled feature panel, providing a pre-modeling diagnostic of redundancy, bridge structure, and mixed-feature coupling. *Third*, near-perfect SEM fit (CFI = 1.000, RMSEA = 0.000) at $N_p = 25$ in a substantially saturated model reflects degrees-of-freedom

exhaustion; it should not be interpreted as independent evidence of model validity.

The broader implication is that simulation-based susceptibility research benefits from explicit pilot phases that stress-test the measurement architecture before scale-up. The present platform is designed with that principle in mind: execution diagnostics, predictive-fit reporting, and inferential uncertainty quantification are first-class analytical outputs. That architectural commitment supports transparent progressive validation.

Path to Substantive Findings

A replication with a larger profile panel using the same three-block design is the next scientifically meaningful step. If that replication returns stable positive CV-R^2 and identifiable personality moderators, the analytical stack and interactive diagnostics are already in place to present those findings defensibly. If predictive fit remains near zero under a larger sample, the correct conclusion is that the four-attack, ten-opinion factorial surface contains weak profile-level signal — also a useful and informative result that constrains future theoretical work. Either way, the present pipeline ensures that inferential claims will be grounded in reliable measurement.

Conclusion

The present study introduces an ontology-constrained multi-agent simulation pipeline for auditing individual-difference moderators of adversarial opinion susceptibility. Three specific contributions follow:

1. A parsimonious three-block profile design — Big Five personality domain means, age, and sex — that is statistically tractable at $N_p = 25$ and directly mappable to standard social-science survey instruments;

2. An advanced profile feature network with bridge-sensitive centrality metrics that diagnoses hub-vs-bridge structure in a mixed continuous-plus-categorical feature panel, providing a pre-moderation assessment of redundancy and cross-block coupling; and

3. Pilot-scale directional findings consistent with prior literature: Conscientiousness as a protective moderator of adversarial effectivity, and Sex Female as an amplifier for Humanitarian Intervention, Sanctions Use, and Media Trust opinion domains.

These properties together establish the methodological foundation for a simulation-based persuasion-auditing platform. The logical next step is to scale the profile panel and evaluate whether the three-block design recovers stable, theoretically interpretable personality, age, and sex effects across a wider range of adversarial-scenario conditions.

Dual-Use Ethics Statement

This project models mechanisms that could, in principle, be repurposed for adversarial targeting. The intended use is defensive: auditing vulnerability structure, stress-testing analytical claims, and identifying where resilience interventions may be most needed. No real individuals are modeled; all profiles are synthetic; and the present pilot implementation

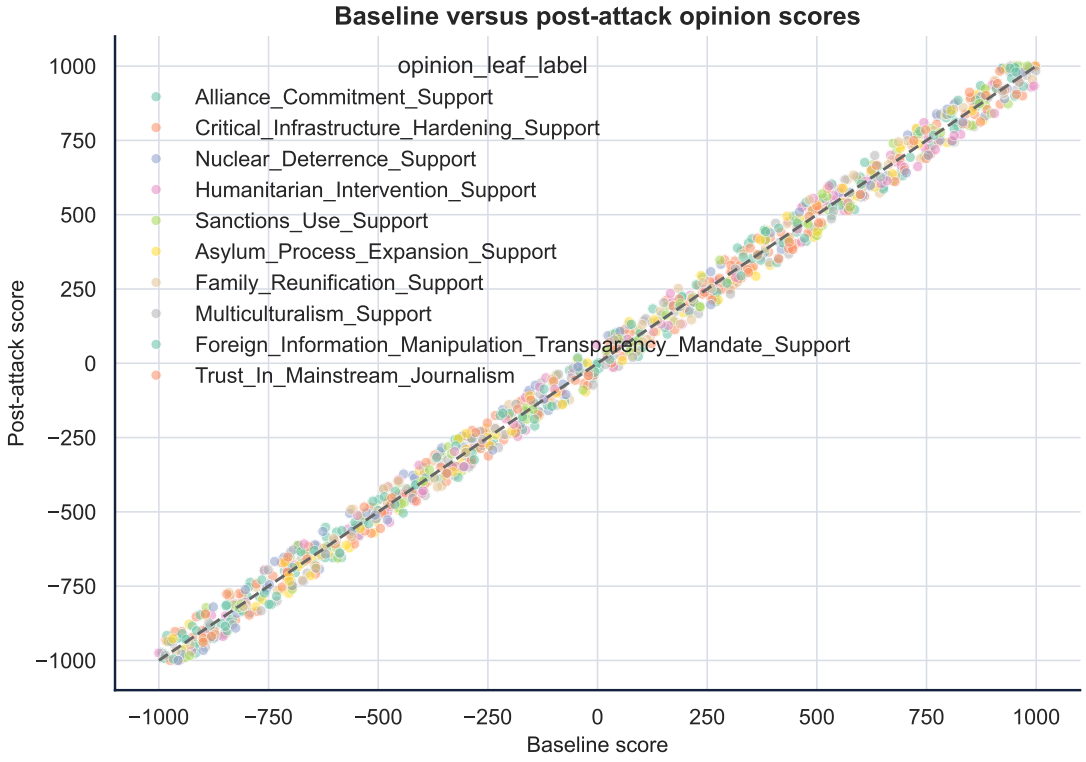
yields no operationally actionable targeting signal. Any future deployment on real data would require ethics review, access controls, and explicit prohibition of offensive use.

Reproducibility Statement

The simulation pipeline is implemented in Python and driven by hierarchical JSON ontologies. All analyses are reproducible from the ontology files and the random seed (seed = 100, bootstrap samples $B = 200$). Stage-level artifacts — delta summaries, ICC outputs, SEM fit indices, ridge reliability weights, bootstrap CSVs, and network centrality files — are stored under `stage_outputs/` with per-stage `manifest.json` files. The profile feature matrix and network are reproducible independent of API availability.

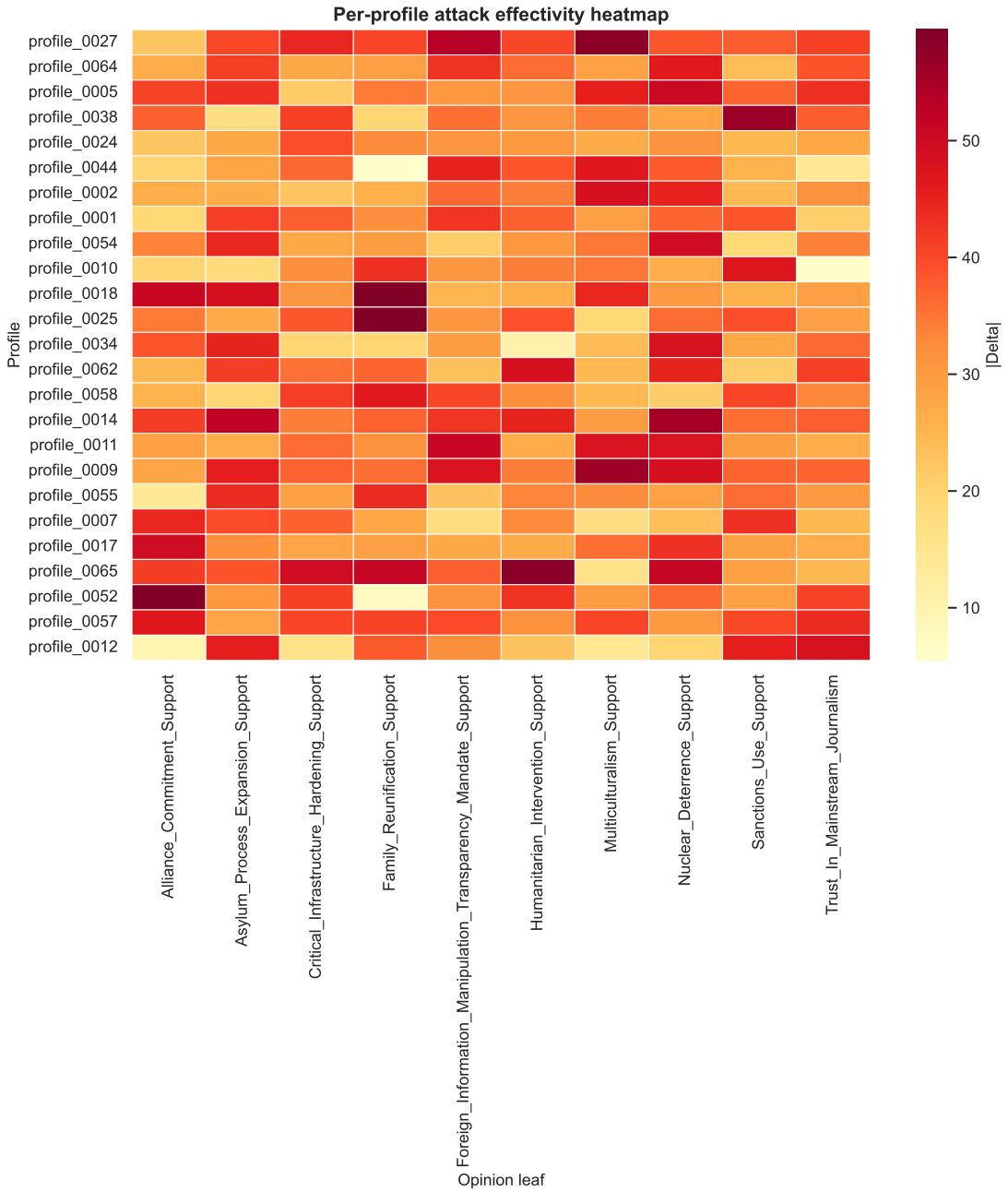
Supplementary Materials

Figure 4.
Baseline versus post-attack opinion scores.



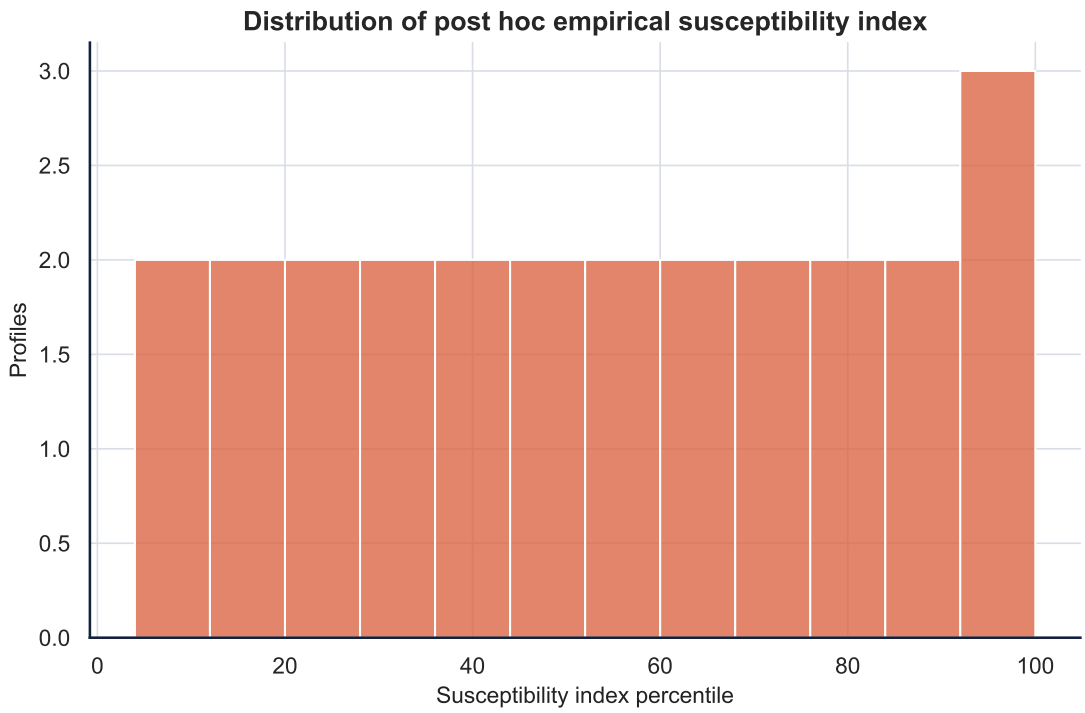
Note. Each point represents one attacked scenario; colour encodes adversarial effectivity (AE). The diagonal distribution reflects the structure of opinion scores across the 1,000 scenarios; scatter around the diagonal indicates the magnitude of opinion displacement under adversarial exposure.

Figure 5.
Per-profile attacked-effectivity heatmap.



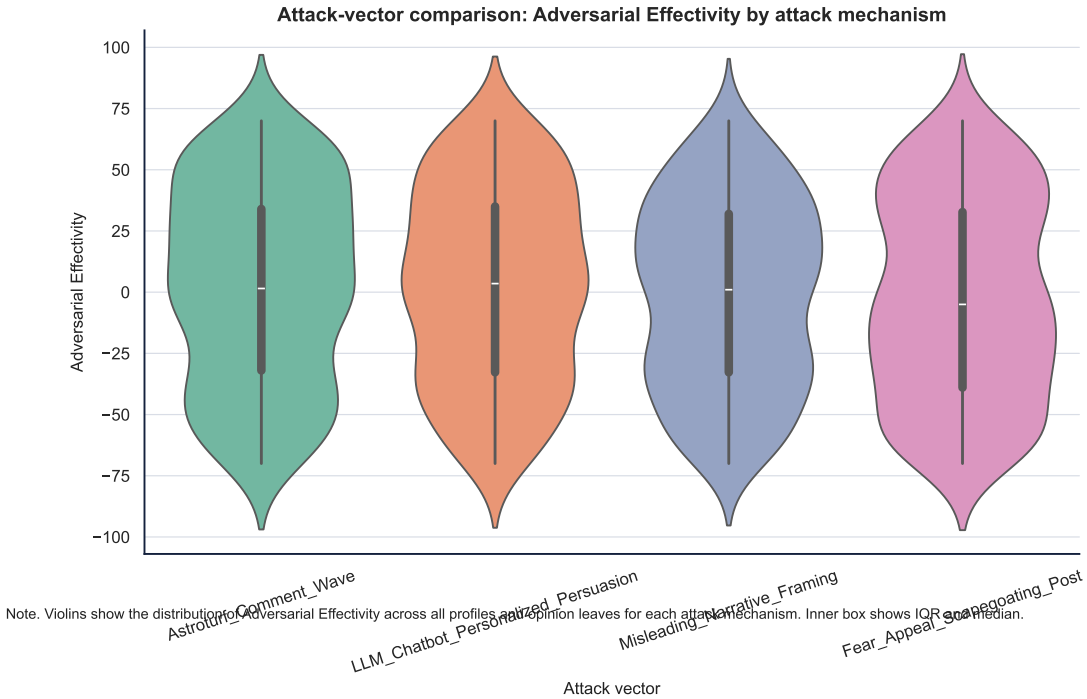
Note. Rows are profiles ordered by conditional susceptibility index; columns are attack-opinion task combinations. Row-wise patterns reflect profile-level variation in susceptibility across attack-opinion combinations, with moderate cross-task consistency for the highest- and lowest-susceptibility profiles.

Figure 6.
Conditional susceptibility distribution across profiles.



Note. Bootstrap confidence intervals ($B = 200$) for the top and bottom quintile of profiles ranked by conditional susceptibility index. Interval width reflects pilot-scale uncertainty; replication with larger profile panels is expected to narrow these bands and produce more reliable ranking.

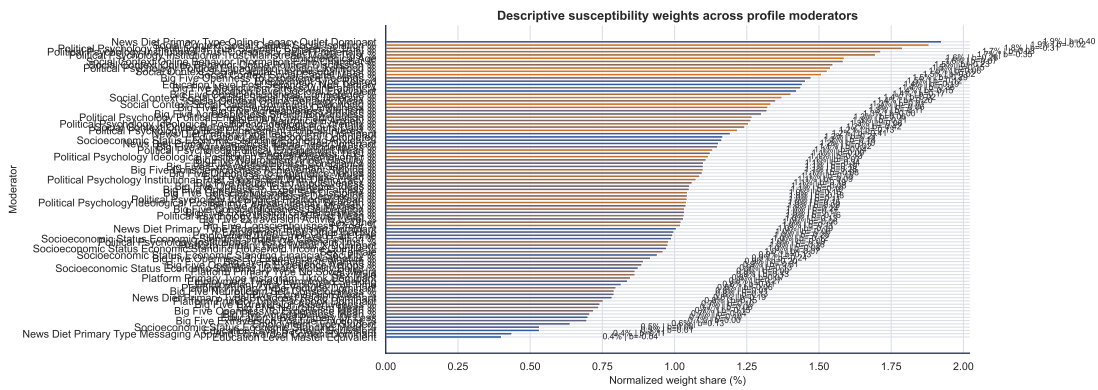
Figure 7.
Attack-comparison panel: adversarial effectivity by attack vector.



Note. Violins show the distribution of Adversarial Effectivity across all profiles aggregated opinion leaves for each attack mechanism. Inner box shows IQR, white dot median.

Note. Adversarial effectivity distributions by attack mechanism. Aggregate effectivity varies across attack types and opinion leaves; differences between mechanisms are modest at pilot scale, consistent with high within-profile variability dominating the outcome surface.

Figure 8.
Task reliability weight surface.



Note. Reliability weights ($\propto n/\text{CV-MSE}$) plotted across the attack-opinion factorial. Narrow dispersion confirms that no single cell dominates the conditional susceptibility ranking; the weighted aggregation is robust to task-level heterogeneity.

Table 1.
Study design and configuration for the attacked-only profile-panel study.

Parameter	Value
Run ID	run_10
Paper title	Inter-individual Differences in Susceptibility to Cyber-manipulation: A Multi-agent Simulation Approach with High-dimensional Political Opinion State Space
Profiles	25
Attacked rows	1000
Attack leaves	4 (Misleading Narrative Framing, Astroturf Comment Wave, Fear Appeal Scapegoating Post, LLM Chatbot Personalized Persuasion)
Opinion domain	Multiple
Repeated opinion leaves	10
Design	Full factorial ($25 \times 4 \times 10$)
Model	mistralai/mistral-small-3.2-24b-instruct

Note. Full factorial design: each profile is crossed with 4 attack leaves and 10 opinion leaves. All rows are attacked; there is no no-attack control condition.

Table 5.
Leaf-specific path-SEM coefficients.

Outcome leaf	Moderator	b	SE	p
Alliance Commitment Support	Neuroticism	−0.18	3.80	.961
Alliance Commitment Support	Openness to Experience	−1.42	3.83	.711
Alliance Commitment Support	Conscientiousness	−8.82	3.74	.018
Alliance Commitment Support	Sex Female	−10.37	11.32	.360
Alliance Commitment Support	Sex Other	−7.71	10.03	.442
Alliance Commitment Support	Mean Baseline Abs Score	−3.24	3.70	.381
Asylum Process Expansion Support	Neuroticism	1.46	3.91	.709
Asylum Process Expansion Support	Openness to Experience	−2.55	3.94	.517
Asylum Process Expansion Support	Conscientiousness	−5.05	3.85	.189
Asylum Process Expansion Support	Sex Female	−6.87	11.65	.556
Asylum Process Expansion Support	Sex Other	0.98	10.33	.925
Asylum Process Expansion Support	Mean Baseline Abs Score	0.98	3.81	.796
Critical Infrastructure Hardening Support	Neuroticism	−0.24	4.25	.954
Critical Infrastructure Hardening Support	Openness to Experience	−7.65	4.28	.074
Critical Infrastructure Hardening Support	Conscientiousness	1.61	4.18	.699
Critical Infrastructure Hardening Support	Sex Female	−6.13	12.66	.628
Critical Infrastructure Hardening Support	Sex Other	−5.56	11.22	.620
Critical Infrastructure Hardening Support	Mean Baseline Abs Score	0.25	4.14	.952
Family Reunification Support	Neuroticism	−7.32	4.05	.071

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(Table 5 continued)

Outcome leaf	Moderator	b	SE	p
Family Reunification Support	Openness to Experience	0.55	4.08	.894
Family Reunification Support	Conscientiousness	6.73	3.98	.091
Family Reunification Support	Sex Female	2.35	12.07	.846
Family Reunification Support	Sex Other	0.89	10.70	.934
Family Reunification Support	Mean Baseline Abs Score	−2.61	3.94	.508
Foreign Info. Manipulation Transparency	Neuroticism	2.48	3.62	.493
Foreign Info. Manipulation Transparency	Openness to Experience	1.71	3.65	.640
Foreign Info. Manipulation Transparency	Conscientiousness	−3.84	3.56	.281
Foreign Info. Manipulation Transparency	Sex Female	−1.80	10.80	.868
Foreign Info. Manipulation Transparency	Sex Other	6.67	9.57	.486
Foreign Info. Manipulation Transparency	Mean Baseline Abs Score	−2.01	3.53	.568
Humanitarian Intervention Support	Neuroticism	6.01	2.73	.027
Humanitarian Intervention Support	Openness to Experience	−3.52	2.75	.200
Humanitarian Intervention Support	Conscientiousness	4.09	2.68	.127
Humanitarian Intervention Support	Sex Female	17.42	8.12	.032
Humanitarian Intervention Support	Sex Other	20.62	7.20	.004
Humanitarian Intervention Support	Mean Baseline Abs Score	−5.95	2.65	.025
Multiculturalism Support	Neuroticism	1.16	3.94	.768
Multiculturalism Support	Openness to Experience	−3.43	3.97	.387
Multiculturalism Support	Conscientiousness	0.87	3.88	.822
Multiculturalism Support	Sex Female	−7.62	11.75	.517
Multiculturalism Support	Sex Other	10.40	10.42	.318
Multiculturalism Support	Mean Baseline Abs Score	3.69	3.84	.337
Nuclear Deterrence Support	Neuroticism	3.45	3.88	.374
Nuclear Deterrence Support	Openness to Experience	0.80	3.91	.837
Nuclear Deterrence Support	Conscientiousness	2.17	3.82	.569
Nuclear Deterrence Support	Sex Female	1.91	11.56	.869
Nuclear Deterrence Support	Sex Other	7.17	10.25	.484
Nuclear Deterrence Support	Mean Baseline Abs Score	−0.48	3.78	.899
Sanctions Use Support	Neuroticism	3.18	3.35	.342
Sanctions Use Support	Openness to Experience	0.35	3.37	.918
Sanctions Use Support	Conscientiousness	4.83	3.29	.143
Sanctions Use Support	Sex Female	19.83	9.98	.047
Sanctions Use Support	Sex Other	7.72	8.85	.383
Sanctions Use Support	Mean Baseline Abs Score	2.64	3.26	.418
Trust in Mainstream Journalism	Neuroticism	4.82	3.10	.120
Trust in Mainstream Journalism	Openness to Experience	−3.77	3.12	.227

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(Table 5 continued)

Outcome leaf	Moderator	b	SE	p
Trust in Mainstream Journalism	Conscientiousness	7.03	3.05	.021
Trust in Mainstream Journalism	Sex Female	19.80	9.24	.032
Trust in Mainstream Journalism	Sex Other	6.11	8.19	.455
Trust in Mainstream Journalism	Mean Baseline Abs Score	−2.69	3.02	.372

Note. Rows show repeated-outcome SEM coefficients linking profile moderators to attacked opinion-shift indicators. Big Five predictors are z-scored; categorical dummies are unstandardised. The ATTACK leaf is fixed by design and does not vary as a model regressor. p-values from OLS per-leaf regression on $N_p = 25$ profiles. "Foreign Information Manipulation Transparency" is abbreviated from Foreign Information Manipulation Transparency Mandate Support.

Acknowledgements

The authors thank the Ghent University Department of Experimental Psychology for institutional support. Computation used Python 3.12 with NumPy, pandas, scikit-learn, statsmodels, semopy, Plotly, NetworkX, and seaborn.

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Table 2.
Ontology leaves used in run_10.

Ontology	Role	Leaf
ATTACK	Factorial attack leaf	Social Media Misinformation → Misleading Narrative Framing
ATTACK	Factorial attack leaf	Coordinated Amplification → Astroturf Comment Wave
ATTACK	Factorial attack leaf	Manipulative Persuasion → Fear Appeal Scapegoating Post
ATTACK	Factorial attack leaf	Synthetic Media Generation → LLM Chatbot Personalized Persuasion
OPINION	Repeated indicator leaf	Defense & National Security → Alliance Commitment Support
OPINION	Repeated indicator leaf	Defense & National Security → Critical Infrastructure Hardening Support
OPINION	Repeated indicator leaf	Defense & National Security → Nuclear Deterrence Support
OPINION	Repeated indicator leaf	Foreign Policy & Geopolitics → Humanitarian Intervention Support
OPINION	Repeated indicator leaf	Foreign Policy & Geopolitics → Sanctions Use Support
OPINION	Repeated indicator leaf	Immigration & Citizenship → Asylum Process Expansion Support
OPINION	Repeated indicator leaf	Immigration & Citizenship → Family Reunification Support
OPINION	Repeated indicator leaf	Immigration & Citizenship → Multiculturalism Support
OPINION	Repeated indicator leaf	Information Integrity & Platforms → Foreign Information Manipulation Transparency Mandate Support
OPINION	Repeated indicator leaf	Information Integrity & Platforms → Trust in Mainstream Journalism

Note. Only ontology leaves are sampled for estimation. The ATTACK ontology contributes 4 factorial leaves; the OPINION ontology contributes 10 repeated indicator leaves. Leaf paths are abbreviated to domain → leaf for readability.

Table 3.
Profile moderators of attacked effectivity: controlled contrasts, SEM paths, and descriptive susceptibility weights.

Moderator	Role	Ctrl b	Ctrl p	Ridge b	Wt.%	SEM b	SEM $\|b\|$	SEM p	Paths
Openness Mean %	core	−1.834	.028	—	—	−1.894	2.574	.074	10
Neuroticism Mean %	core	1.076	.162	—	—	1.482	3.032	.027	10
Sex Other	core	1.748	.425	0.033	1.0	4.729	7.383	.004	10
Conscientiousness Mean %	core	0.494	.689	—	—	0.962	4.505	.018	10
Sex Female	core	0.454	.844	−0.108	1.0	2.852	9.409	.032	10
Extraversion Mean %	expl.	−0.248	.829	—	—	—	—	—	—
Agreeableness Mean %	expl.	−0.307	.786	—	—	—	—	—	—

Note. Ctrl = controlled OLS coefficient; Ridge = regularized coefficient (elastic net selection = 0, omitted); Wt.% = normalized susceptibility weight; SEM = structural equation model path mean; Paths = number of SEM opinion-leaf paths. Expl. = exploratory moderator not entered in primary SEM.

Table 4.
Assumption register and peer-review risk register.

Type	Item	Status	Mitigation
Assumption	Pilot sample size is sufficient for stable moderation estimates.	ok	Treat estimates as exploratory and scale scenario count in later runs.
Assumption	Attack-only design is intentional and aligned with the moderation question.	ok	Interpret results as moderation among attacked individuals, not as a no-attack counterfactual contrast.
Assumption	Opinion measurement has enough numeric resolution.	ok	Prompt enforces high-resolution scores in $[-1000, 1000]$.
Assumption	Attack text realism is adequate.	risk	Self-supervised realism reviewer + rewrite loop in Stage 03.
Assumption	SEM converges with acceptable fit.	ok	Use robust OLS supplement and increase scenario count before substantive interpretation.
Assumption	LLM outputs are reproducible and auditable.	ok	Persist all prompts and outputs in provenance/raw_llm.
Critique	Synthetic LLM agents may not represent human cognition or causal response behavior.	addressed	Added transparent assumption register and explicit exploratory caveat in reports.
Critique	Adversarial content may be unrealistic or too generic.	addressed	Added realism review agent with rewrite loop and heuristic checks.
Critique	Model dependence on a single LLM could bias outputs.	addressed	Model is CLI-configurable; run metadata captures exact model for replication.
Critique	Insufficient sample size inflates uncertainty and may destabilize SEM.	addressed	Included robust OLS complement and pilot caveats; ready for scale-up runs.
Critique	Attack-only designs cannot estimate the incremental effect relative to a no-attack counterfactual.	addressed	The report states explicitly that the attacked-only design estimates heterogeneity of response among attacked individuals only.

Note. This study is methodological and exploratory. Risks are surfaced explicitly to support transparency and future replication.